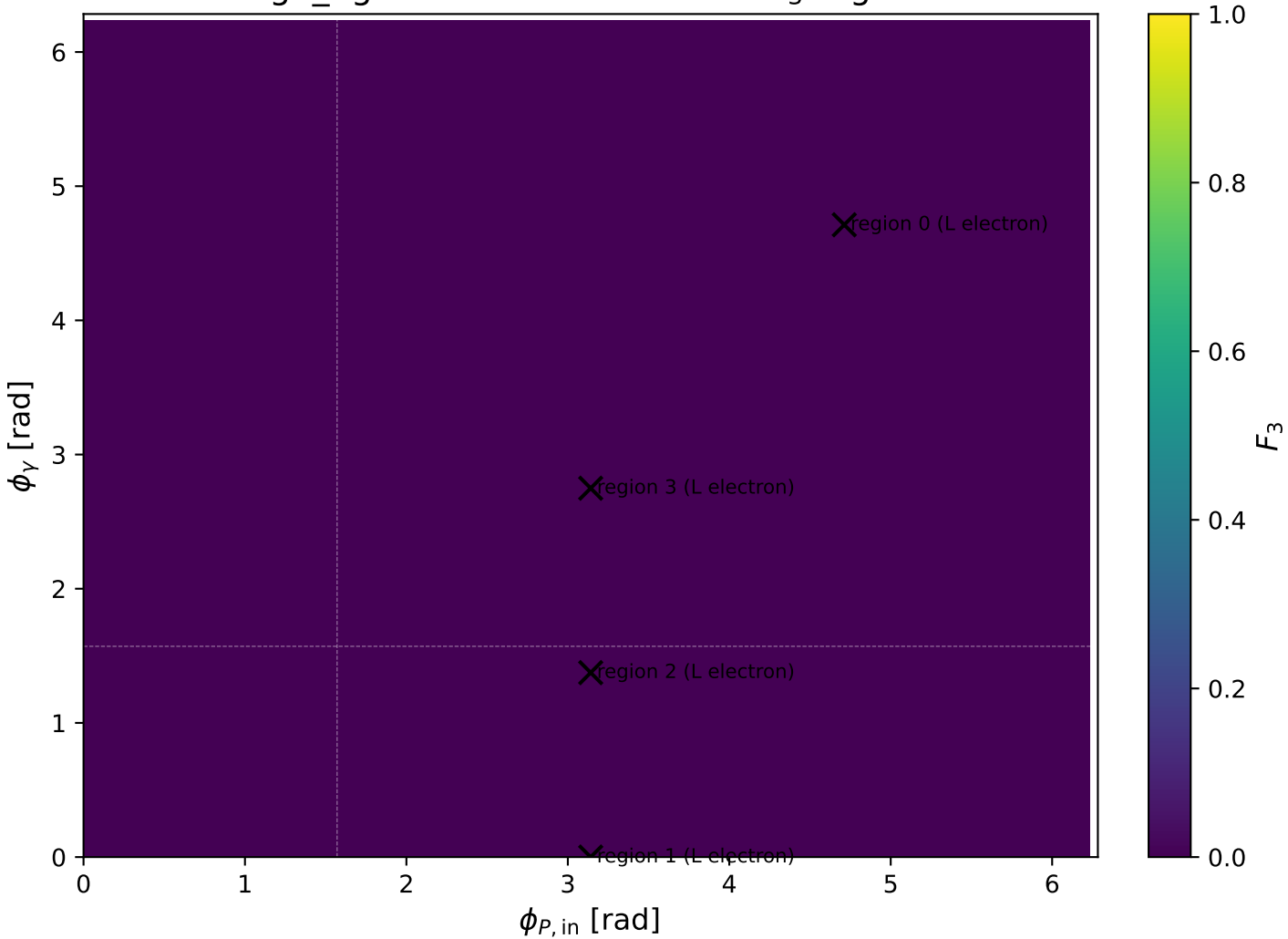
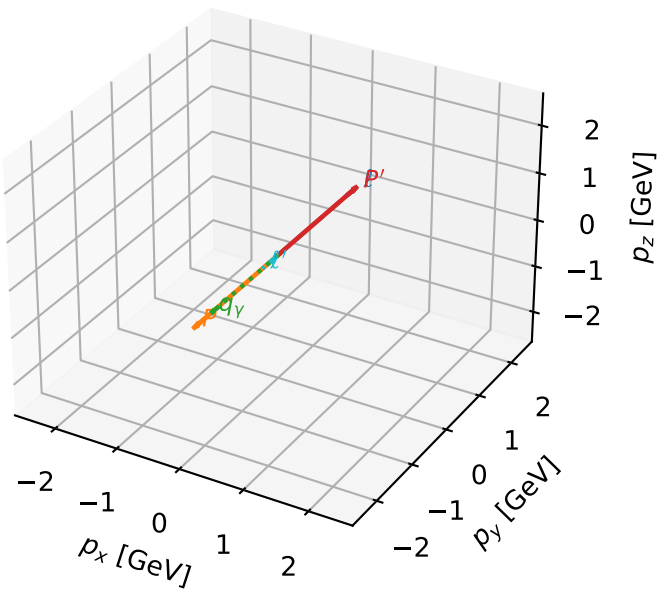


high\_Egamma L electron: max  $F_3$  regions



# L electron characteristic max $F_3$ configuration

3D momenta

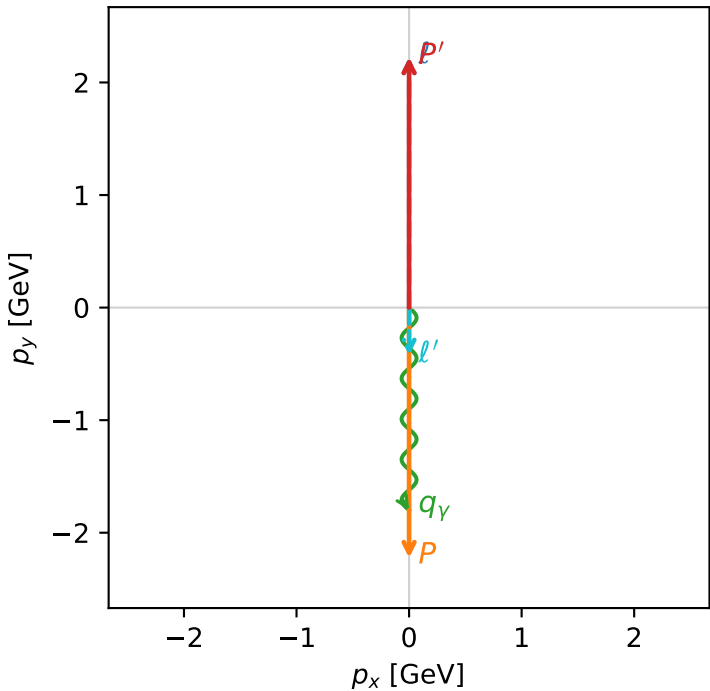


f3\_high\_egamma\_l\_electron\_region\_0  $F_3=1.89386\text{e-}14$   
region: high\_Egamma\_L\_electron\_region\_0  
outgoing-state purity: 1  
kinematic point: high\_s\_high\_theta\_in\_high\_Egamma  
incoming state:  $\frac{1}{2}\sum_{h_p}\rho(L_e, h_p)$

$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{p}|=2.22442$ ,  $|\vec{p}'|=2.22442$   
 $\theta_{\text{in}}=1.57079$ ,  $\phi_l=1.5708$ ,  $\phi_p=4.71239$   
 $E_V=1.8$ ,  $\phi_V=4.71239$   
 $Q^2=3.77633$ ,  $x_B=0.184436$ ,  $t=-19.7921$ ,  $W^2=17.5785$ ,  $y=0.992198$   
 $\theta(l', \gamma)=0$  deg

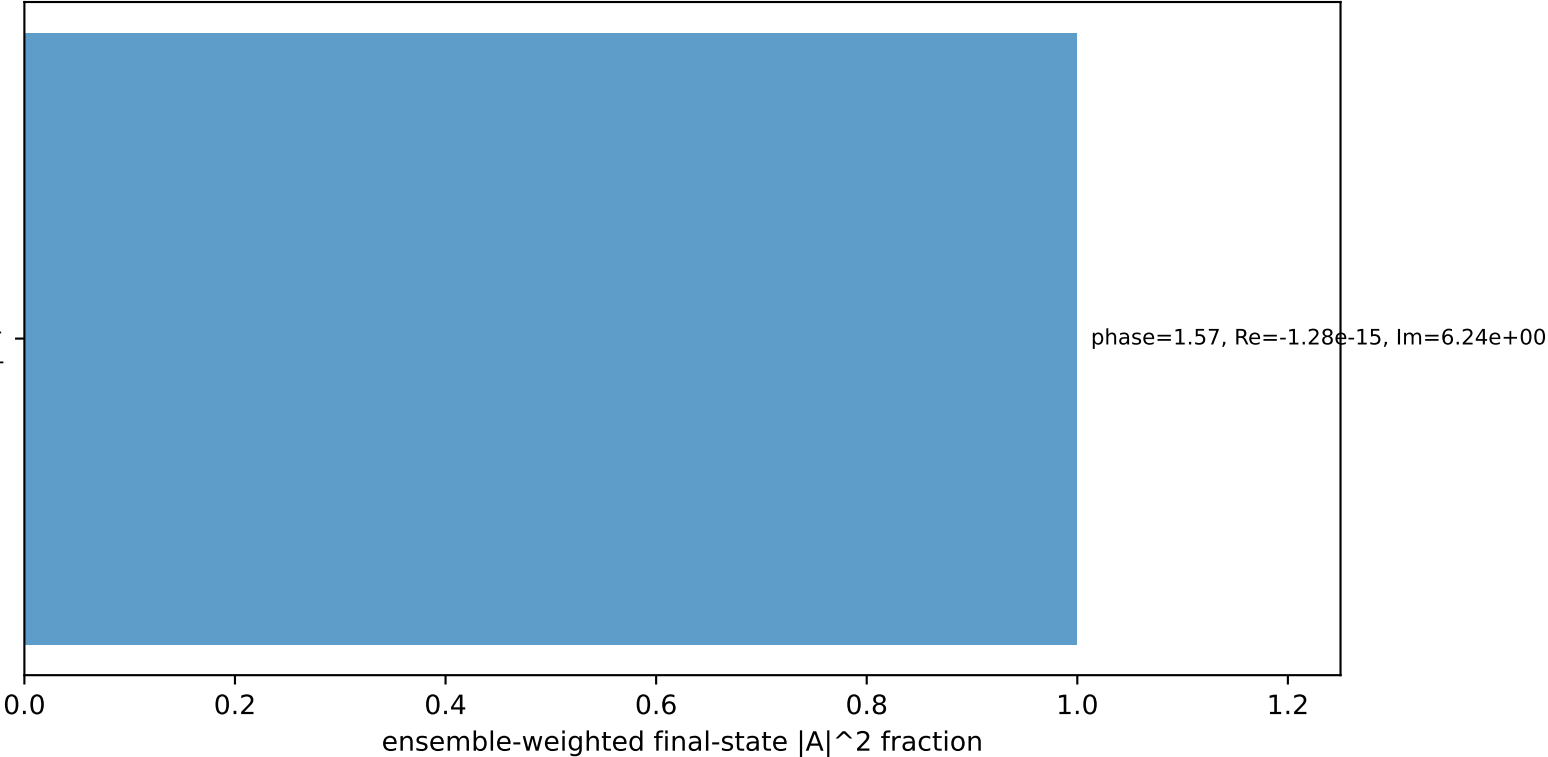
four-momenta  $[E, p_x, p_y, p_z]$  GeV:  
 $l$   $E= 2.2244$   $p_x= 0.0000$   $p_y= 2.2244$   $p_z= -0.0000$   
 $P$   $E= 2.4141$   $p_x= -0.0000$   $p_y= -2.2244$   $p_z= 0.0000$   
 $l'$   $E= 0.4244$   $p_x= 0.0000$   $p_y= -0.4244$   $p_z= 0.0000$   
 $P'$   $E= 2.4141$   $p_x= 0.0000$   $p_y= 2.2244$   $p_z= 0.0000$   
 $q_Y$   $E= 1.8000$   $p_x= -0.0000$   $p_y= -1.8000$   $p_z= 0.0000$

Transverse plane



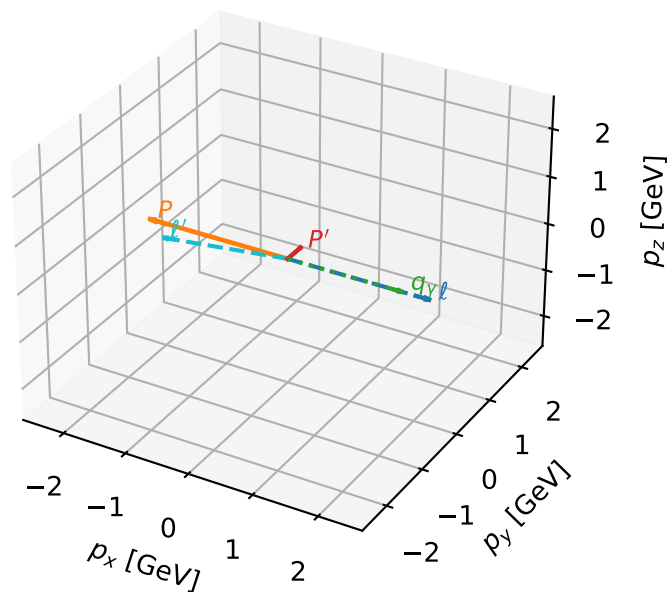
$h_e=-1$ ,  $h_p=+1$ ,  $h_\gamma=+1$   
electron L, proton h=+1

Leading initial-ensemble/final-state components (N=1, retained=100.0%)



# L electron characteristic max $F_3$ configuration

3D momenta



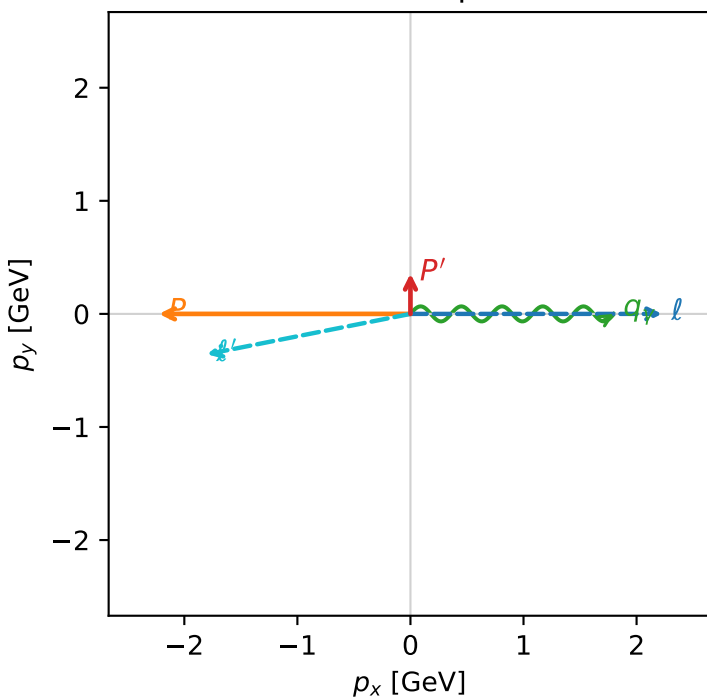
f3\_high\_egamma\_l\_electron\_region\_1 F\_3=0  
region: high\_Egamma\_L\_electron\_region\_1  
outgoing-state purity: 0.997964  
kinematic point: high\_s\_high\_theta\_in\_high\_Egamma  
incoming state:  $\frac{1}{2} \sum_{h_p} \rho(L_e, h_p)$

$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22442$ ,  $|\vec{P}'|=0.356666$   
 $\theta_{in}=1.57079$ ,  $\phi_l=0$ ,  $\phi_P=3.14159$   
 $E_\gamma=1.8$ ,  $\phi_\gamma=0$   
 $Q^2=16.1715$ ,  $x_B=0.817396$ ,  $t=-3.08551$ ,  $W^2=4.49252$ ,  $y=0.958721$   
 $\theta(l', \gamma)=168.792$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:

|       |      |        |        |         |        |         |        |         |
|-------|------|--------|--------|---------|--------|---------|--------|---------|
| $l$   | $E=$ | 2.2244 | $p_x=$ | 2.2244  | $p_y=$ | -0.0000 | $p_z=$ | -0.0000 |
| $P$   | $E=$ | 2.4141 | $p_x=$ | -2.2244 | $p_y=$ | 0.0000  | $p_z=$ | 0.0000  |
| $l'$  | $E=$ | 1.8350 | $p_x=$ | -1.8000 | $p_y=$ | -0.3567 | $p_z=$ | 0.0000  |
| $P'$  | $E=$ | 1.0035 | $p_x=$ | 0.0000  | $p_y=$ | 0.3567  | $p_z=$ | 0.0000  |
| $q_Y$ | $E=$ | 1.8000 | $p_x=$ | 1.8000  | $p_y=$ | 0.0000  | $p_z=$ | 0.0000  |

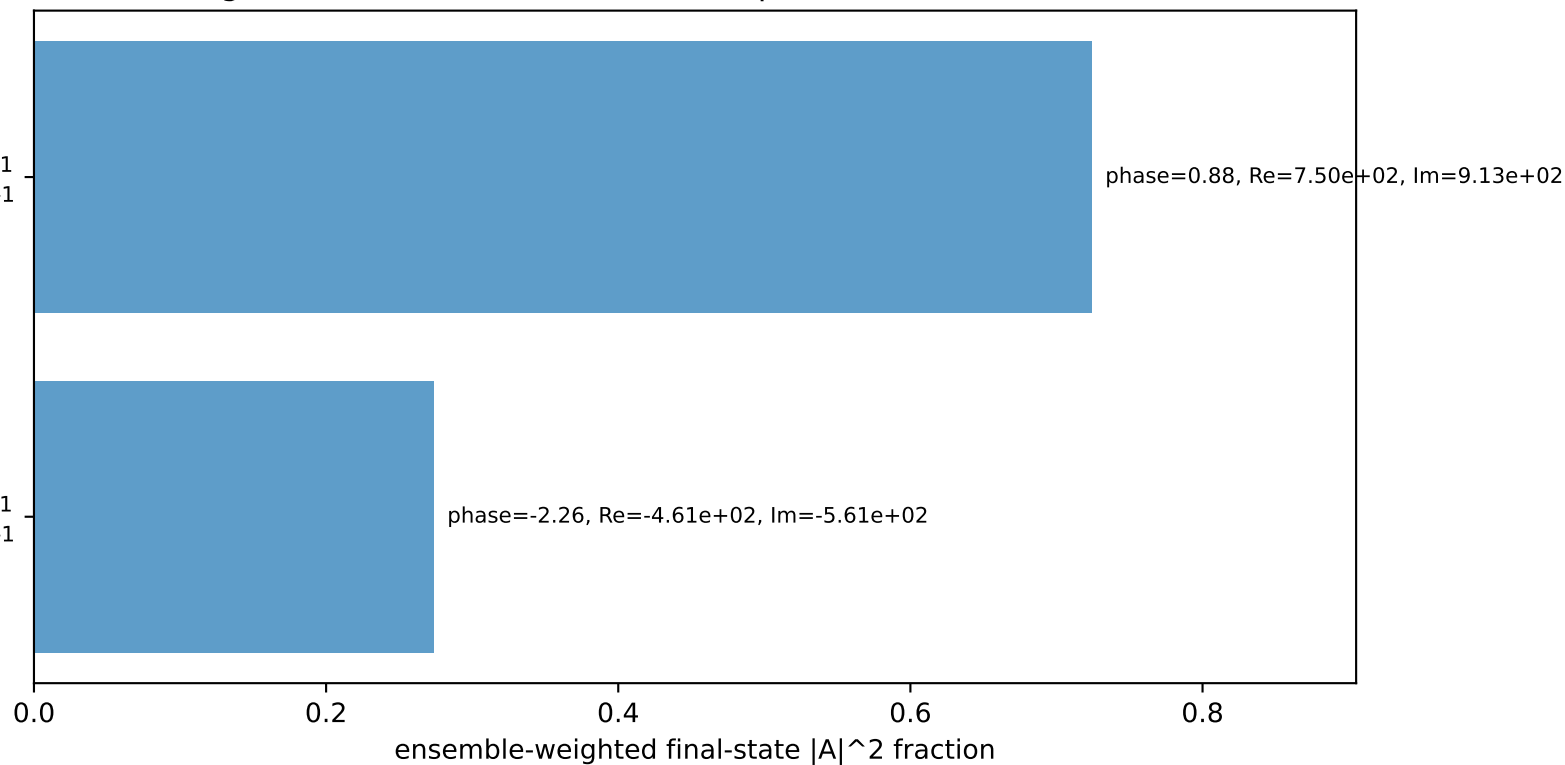
Transverse plane



$h_e=-1, h_p=-1, h_\gamma=+1$   
electron L, proton h=-1

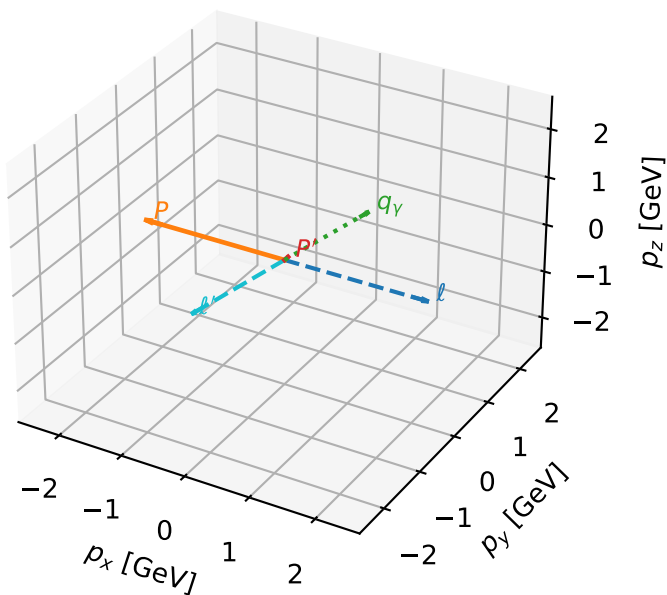
$h_e=-1, h_p=+1, h_\gamma=+1$   
electron L, proton h=-1

Leading initial-ensemble/final-state components (N=2, retained=99.8%)



# L electron characteristic max $F_3$ configuration

3D momenta

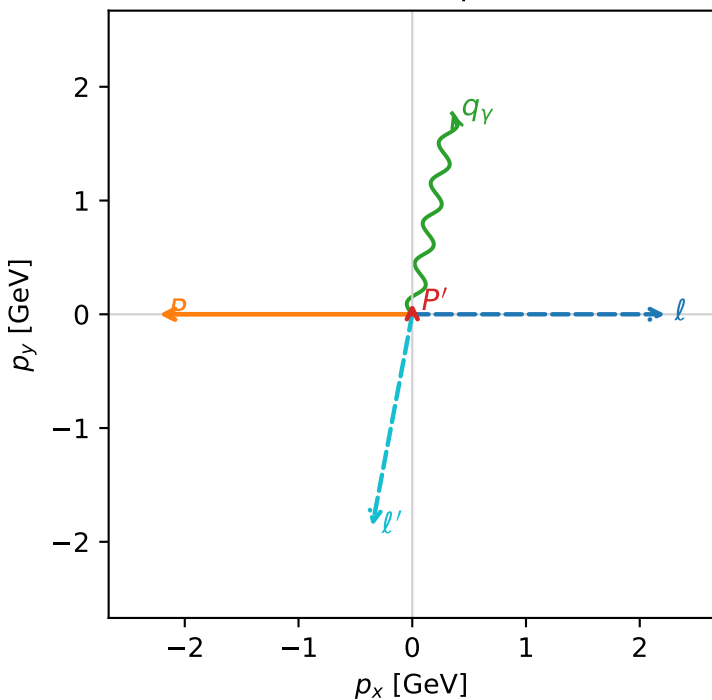


f3\_high\_egamma\_l\_electron\_region\_2 F\_3=0  
 region: high\_Egamma\_L\_electron\_region\_2  
 outgoing-state purity: 0.810065  
 kinematic point: high\_s\_high\_theta\_in\_high\_Egamma  
 incoming state:  $\frac{1}{2} \sum_{h_p} \rho(L_e, h_p)$

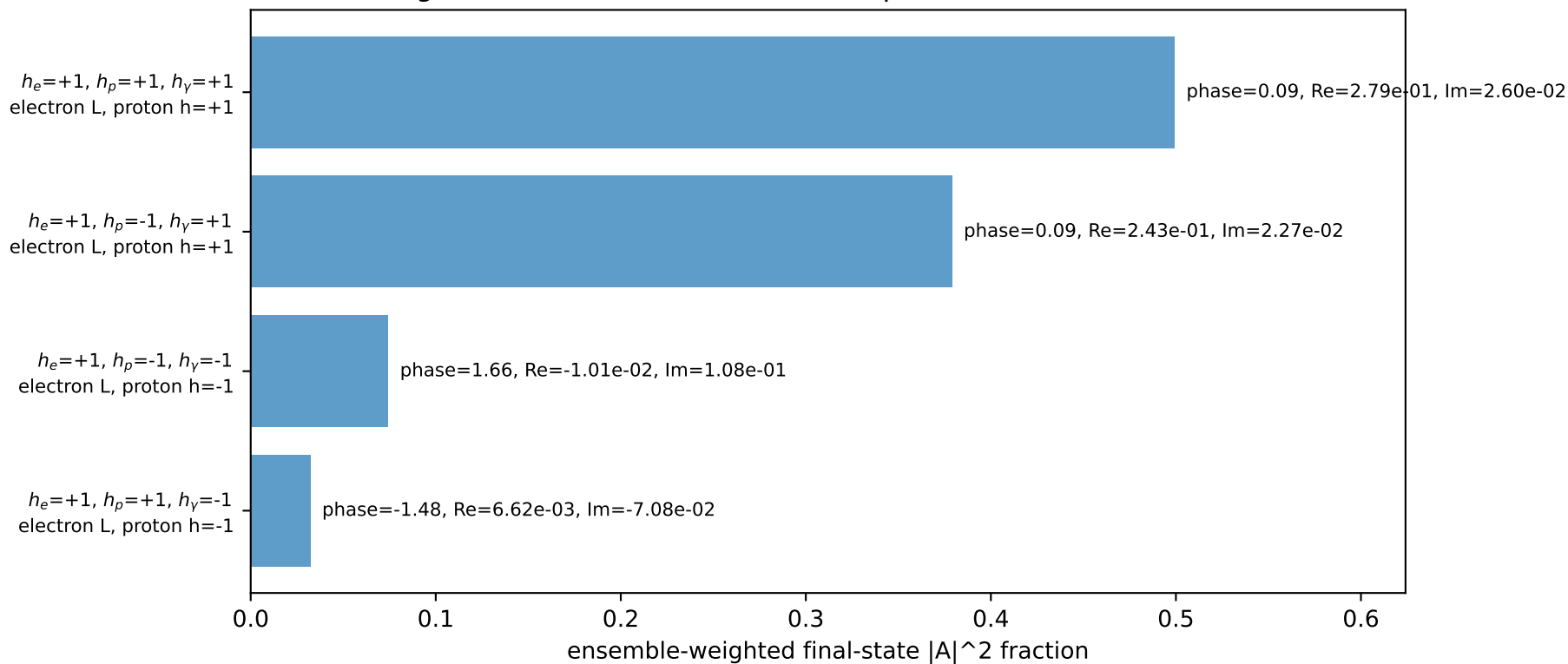
$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22442$ ,  $|\vec{P}'|=0.0972621$   
 $\theta_{in}=1.57079$ ,  $\phi_l=0$ ,  $\phi_p=3.14159$   
 $E_\gamma=1.8$ ,  $\phi_\gamma=1.37445$   
 $Q^2=9.99498$ ,  $x_B=0.766106$ ,  $t=-2.79344$ ,  $W^2=3.93133$ ,  $y=0.632219$   
 $\theta(l', \gamma)=179.426$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:  
 $l$   $E= 2.2244$   $p_x= 2.2244$   $p_y= -0.0000$   $p_z= -0.0000$   
 $P$   $E= 2.4141$   $p_x= -2.2244$   $p_y= 0.0000$   $p_z= 0.0000$   
 $l'$   $E= 1.8955$   $p_x= -0.3512$   $p_y= -1.8627$   $p_z= 0.0000$   
 $P'$   $E= 0.9430$   $p_x= 0.0000$   $p_y= 0.0973$   $p_z= 0.0000$   
 $q_\gamma$   $E= 1.8000$   $p_x= 0.3512$   $p_y= 1.7654$   $p_z= 0.0000$

Transverse plane

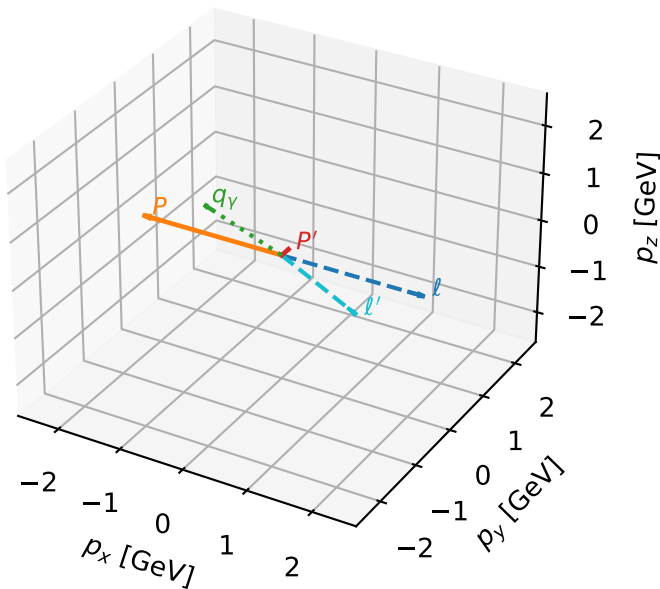


Leading initial-ensemble/final-state components (N=4, retained=98.5%)



# L electron characteristic max $F_3$ configuration

3D momenta

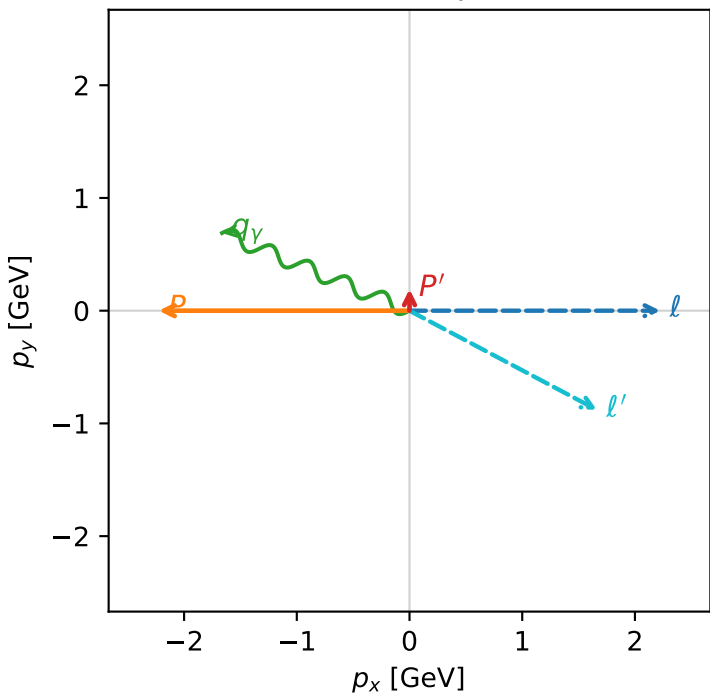


f3\_high\_egamma\_l\_electron\_region\_3 F\_3=0  
region: high\_Egamma L\_electron\_region\_3  
outgoing-state purity: 0.525883  
kinematic point: high\_s\_high\_theta\_in\_high\_Egamma  
incoming state:  $\frac{1}{2} \sum_{h_p} \rho(L_e, h_p)$

$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22442$ ,  $|\vec{P}'|=0.190824$   
 $\theta_{in}=1.57079$ ,  $\phi_l=0$ ,  $\phi_p=3.14159$   
 $E_V=1.8$ ,  $\phi_V=2.74889$   
 $Q^2=0.971273$ ,  $x_B=0.233797$ ,  $t=-2.86193$ ,  $W^2=4.06292$ ,  $y=0.201316$   
 $\theta(l', \gamma)=174.623$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:  
 $l$   $E= 2.2244$   $p_x= 2.2244$   $p_y= -0.0000$   $p_z= -0.0000$   
 $P$   $E= 2.4141$   $p_x= -2.2244$   $p_y= 0.0000$   $p_z= 0.0000$   
 $l'$   $E= 1.8813$   $p_x= 1.6630$   $p_y= -0.8797$   $p_z= 0.0000$   
 $P'$   $E= 0.9572$   $p_x= 0.0000$   $p_y= 0.1908$   $p_z= 0.0000$   
 $q_\gamma$   $E= 1.8000$   $p_x= -1.6630$   $p_y= 0.6888$   $p_z= 0.0000$

Transverse plane



Leading initial-ensemble/final-state components (N=4, retained=99.9%)

